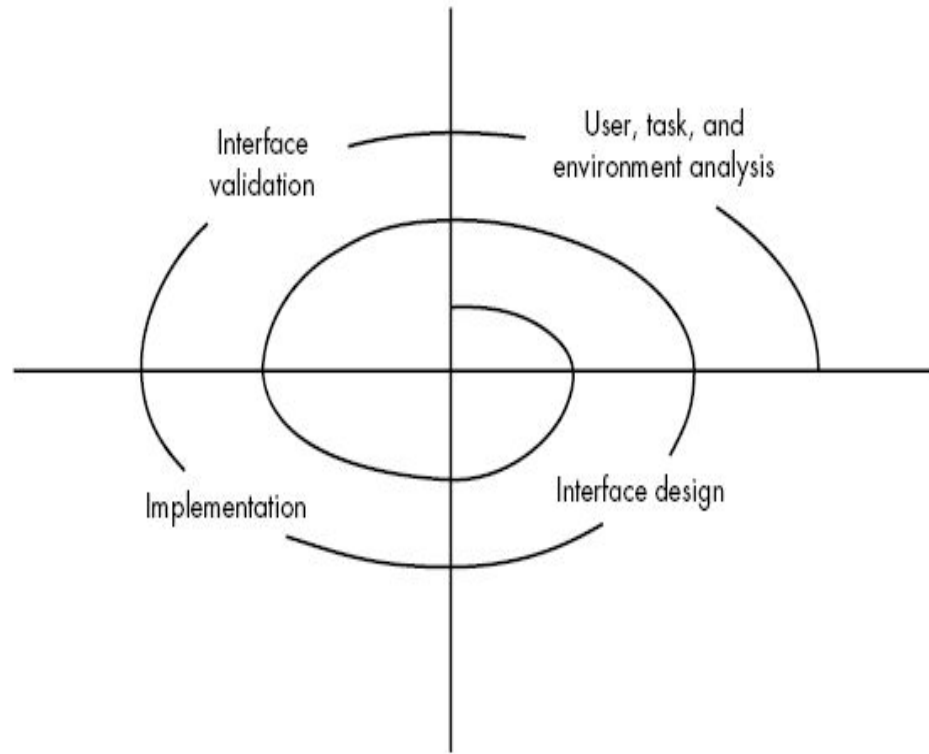
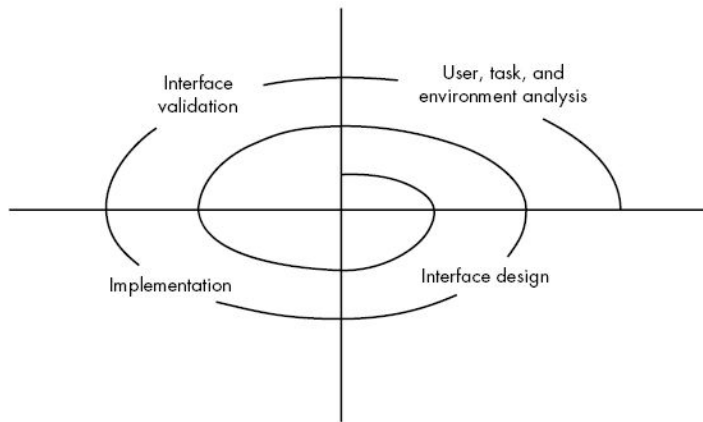


UI analysis & design Process



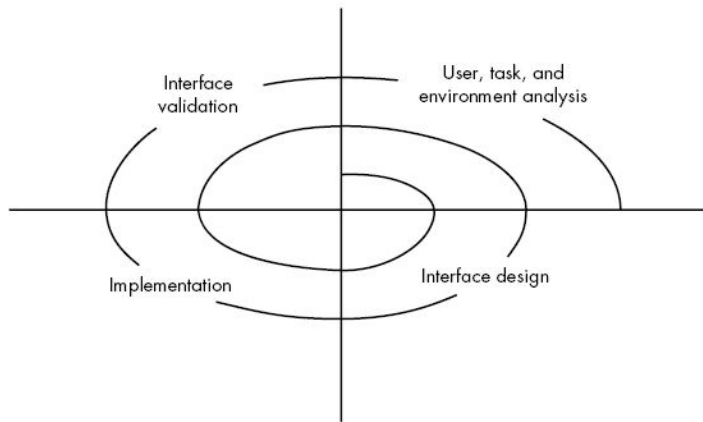
- Iterative process – It uses the spiral process model
- Four steps are:
 - Interface analysis
 - Interface design
 - Interface implementation/construction involves a prototyping approach
 - Interface validation
- Since it's spiral model so these steps need to be processed again and again for design betterment.

UI analysis & design Process



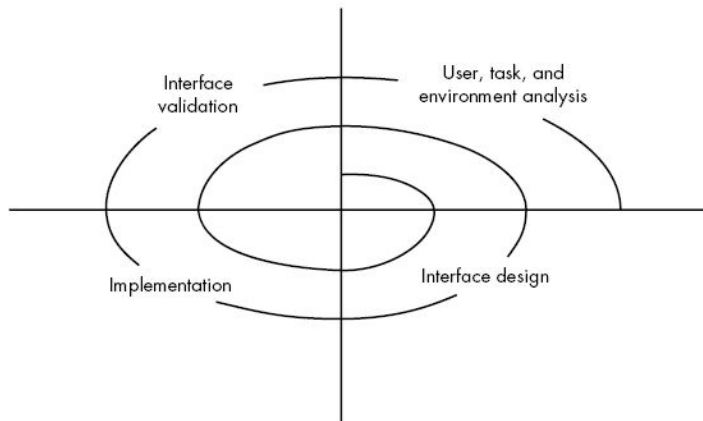
- User analysis:
 - who are going to use the system?
 - What is the skill level, business perception ?
 - User diversity?
 - Gather requirements for each user category & understand system by each category perspective.

UI analysis & design Process



- Interface design:
 - A set of interface objects and actions along with their screen representation that allows user to perform all tasks and fulfil the usability goals of the system.
- Interface Construction/implementation:
 - Create prototype against a use case scenarios.

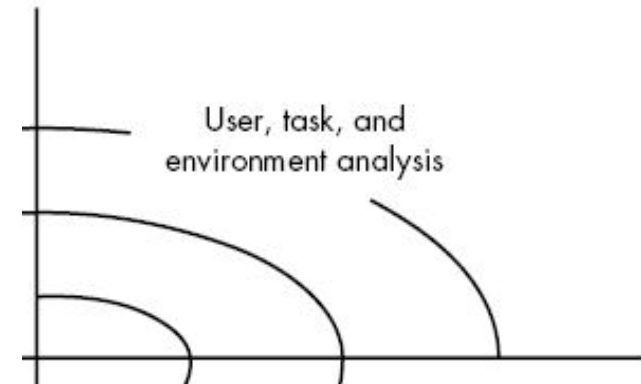
UI analysis & design Process



- Interface validation:
 - Check if the system accommodates all variety of tasks?
 - Whether the system is easy to use & learn?
 - users' acceptance of a system as useful tool
- Now, we'll cover each separately in detail

Interface Analysis

- User Analysis:
- Understand problem before designing the solution.
 - Understand users of the system,
 - the system goals (tasks),
 - the displayed content,
 - the environment in which tasks is conducted.
- The mental image and the design model must converge.
 - Can be achieved by understanding the users and how they use the system



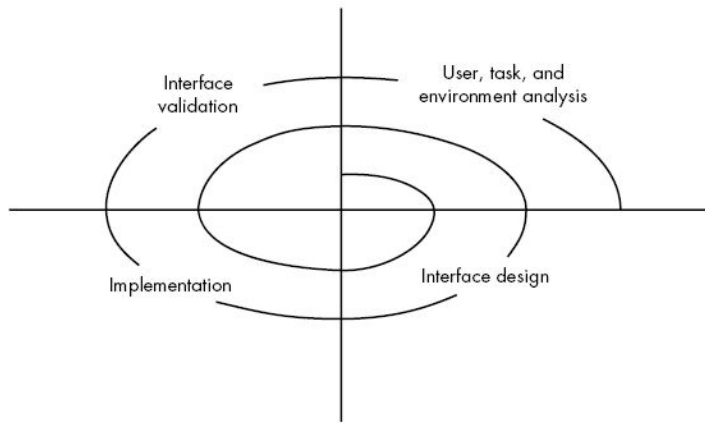
Interface Analysis

- Information for system understanding can be obtained by:
 - *User interviews*
 - development team members met with end users to identify their needs and work culture.
 - *Marketing & sales input*
 - Take the market survey and check in how many ways the product could come handy.
 - *Support input*
 - Support staff takes feedback of the what work or what not. Which features are easy to use or not etc.

User Analysis questions for better user categorization

- Are **users trained professionals, technician, clerical, or manufacturing workers** ?
- **Level of formal education** of an average user?
- Are the users capable of learning from written materials or have they expressed a desire for classroom training?
- users **expert typists / keyboard phobic** ?
- **age range** of the user community?
- User's **gender**?
- How are users compensated for the work they perform?
- Working hours: **normal office hours / until the job is done** ?
- **Software usage frequency** ? (mostly/occasionally)
- **primary spoken language among users** ?
- **consequences of a user's mistake while using the system** ?
- Are users experts in the subject matter that is addressed by the system?
- Are users interested in learning about the technology that sits behind the interface?

UI analysis & design Process



- task analysis:
 - Detailed task analysis done after user analysis.
 - The tasks that users will perform to accomplish system goals.
- User environment analysis:
 - done to check if the current physical environment be able to cater the system needs?
- Output: Analysis model for interface

Interface Analysis

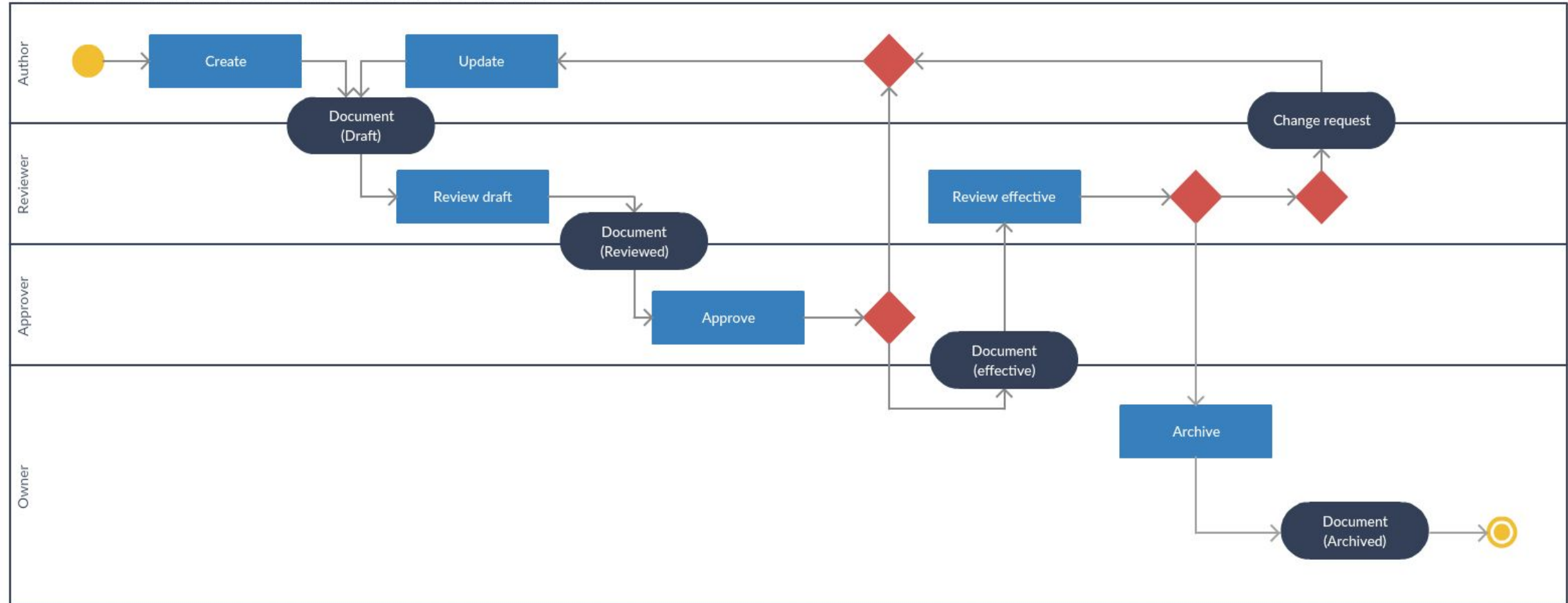
- Task Analysis & Modelling
- answer the given questions ...
 - Task identification in specific circumstances.
 - Tasks and sub tasks performed by user
 - sequence of work tasks—the workflow
- Use-cases:
 - define basic interaction of actor and system
 - written using informal paragraphs.
 - Can be used to derive tasks, sub tasks and interfaces
- Task elaboration:
 - Stepwise elaboration or task refinement of interactive tasks
 - Derive either manually or use a preexisting system to identify them
- Object elaboration:
 - extract physical objects from system to define their classes and behavior
- Workflow analysis:
 - How defines how a work process is
 - a work process is completed when several people (and roles) are involved
 - Shown using UML swim line diagram

Interface Analysis

- Task Analysis & Modelling

Swim line diagram

DOCUMENT MANAGEMENT SYSTEM for ABC Co.

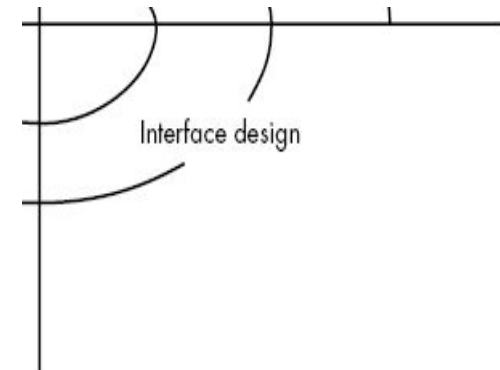


Interface Analysis

- Analysis of Display Content
 - Format of the content
 - Aesthetics
 - Content displayed using stepwise refinement approach
- Analysis of Physical Work Environment
 - Work environment should be helpful in proper operation and concentration .
 - Must co-relate with the designed software aesthetics.

Interface Design Steps

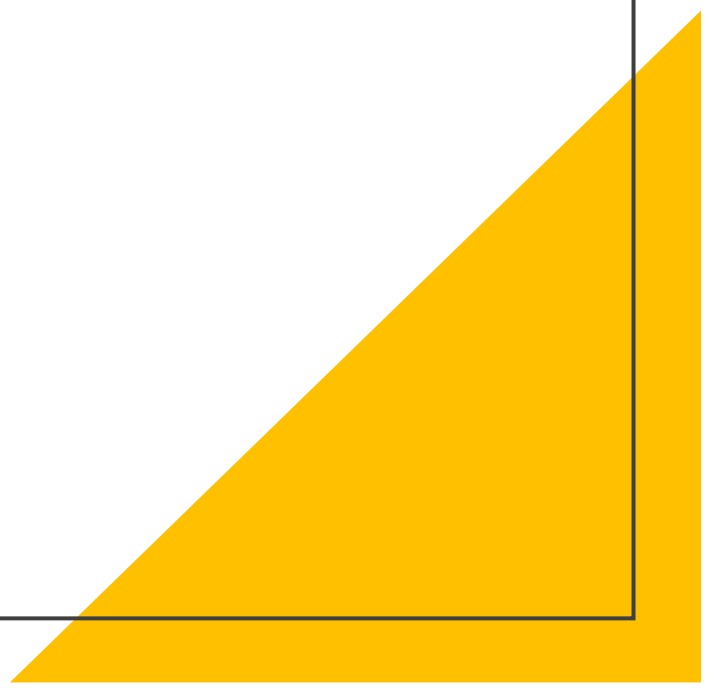
- UI Design models state that:
 - Using information developed during interface analysis,
 - define interface objects (buttons, text fields, video etc.) and operations (tasks)
 - Define events (user actions) that will change the state of the UI.
 - Depict each interface state as it will actually look to the end-user. (using sketches).
 - Show how the user infer the System state from information provided through the interface.



Interface Design Steps

- Apply interface design steps
 - Define the use-case:
 - Create the list of objects and actions
 - These objects & actions are categorized into different type.
 - Identify which are source (e.g., printer icon), target (e.g., paper) and application objects(e.g., screen, entity/db table).
- Use UI design patterns
 - As specific design pattern is formulated for a well bounded solution.
 - E.g., **for a calendar UI**, a Calendar Strip is created that allows to traverse in future dates, marking the current one.

Interface Design Steps

- Design Issues
 - Some of the issues in interactive systems are:
 - System Response time
 - Help facilities
 - Error handling
 - Menu & command labelling
- 
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Interface Design Steps

- Design Issue 1: System Response time
 - Primary complaint for almost every interactive system
 - Two characteristics:
 - Length: longer response times are not appreciated, led to user frustration
 - Variability: varying response time led to user confusion, like whether something went wrong at backend which is leading to a different response time than average response time.

Interface Design Steps

- Design Issue 2: Help Facilitates

- Online helpdesk
- Detailed user manuals
- Design issues while providing Help facilities

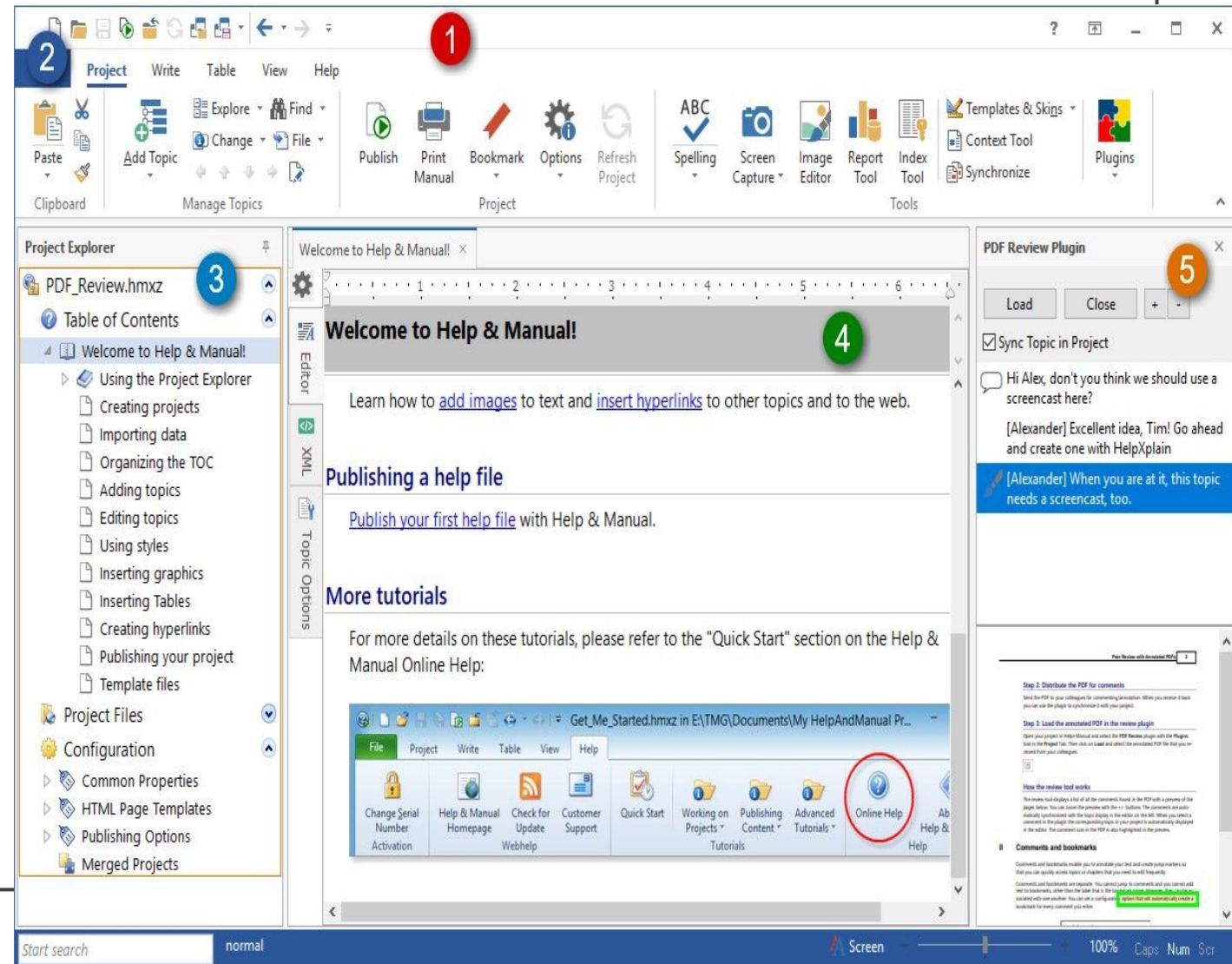
- **Help availability** for all functionalities?

- **How will user demand help?**

(give menu for that or a function key mapped to that)

- **How is help presented?** (via another window or a printed manual)

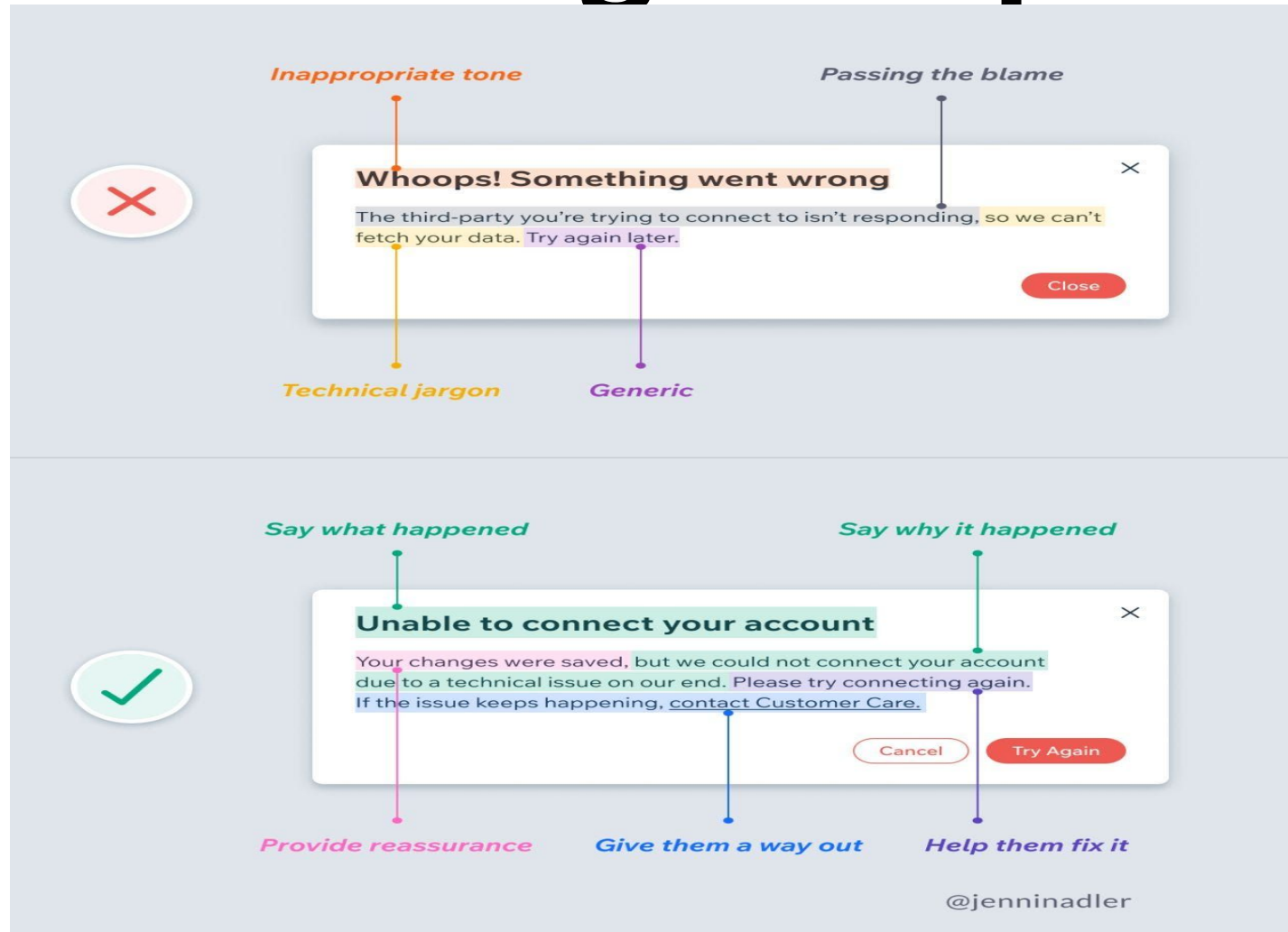
- **How to return to normal UI after consulting help**



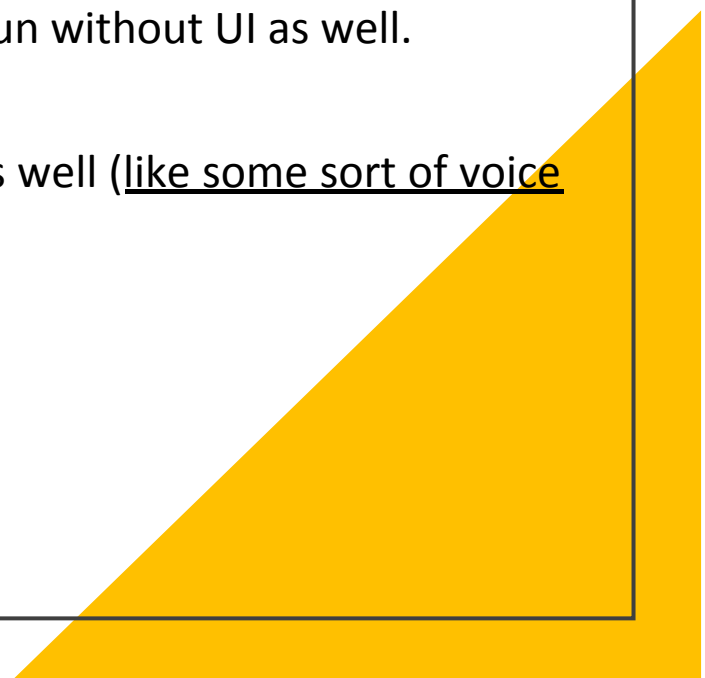
Interface Design Steps

- Design Issue 3: Error Handling
 - Error messages & warnings are bad news to users when something went wrong.
 - Error message must be **explanatory** and should **include some remedies** for that (via hyperlinks)
 - A **good error message** includes:
 - Describe problem simply
 - Provide valuable advice
 - List the negative consequences of the error
 - Message must prompt visually using a color scheme that easily categorize it as an error.

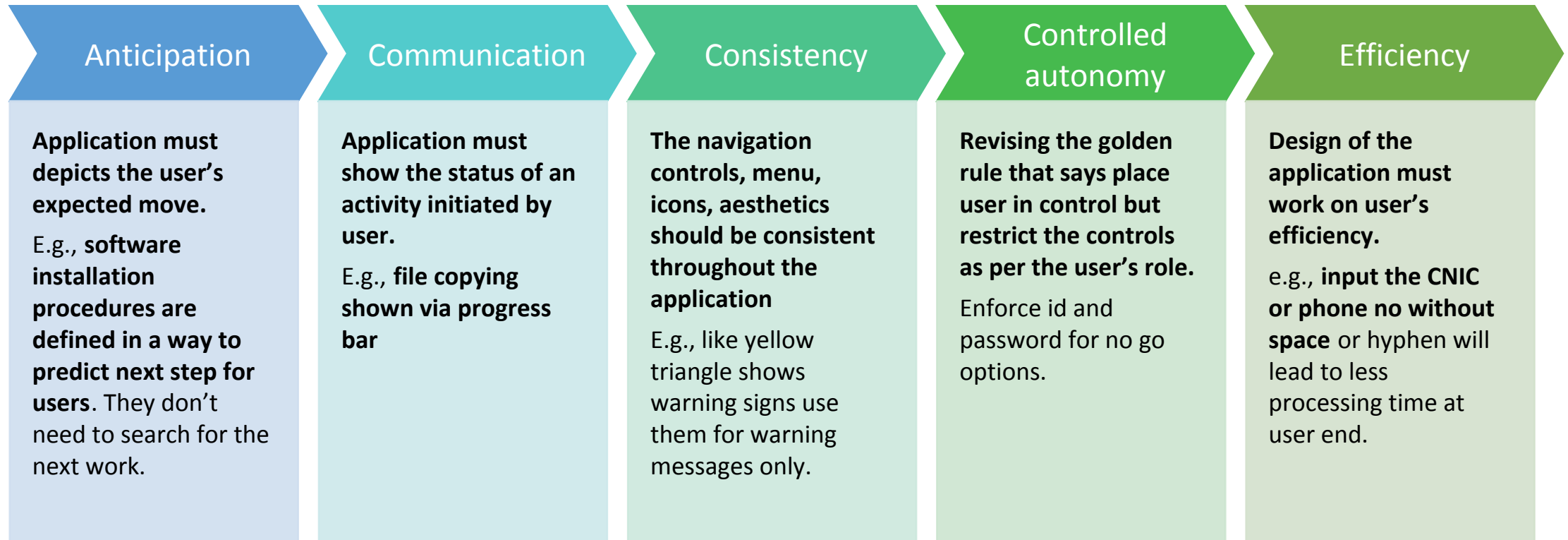
Interface Design Steps



Interface Design Steps

- Design Issue 4: Menu & Command labeling
 - Different styles like command oriented arch and UI based arch.
 - Both are used frequently
 - How commands are assigned to each menu item so that it may run without UI as well.
 - Design Issue 5: Application Accessibility
 - UI should cater the needs of the physically challenged persons as well (like some sort of voice inputs could be accepted to facilitate handicapped)
 - Design Issue 6: internationalization
 - UI must target the global market
 - Follow global design standards & facilitate multiple languages.
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Revised Interface Design Guidelines



Revised Interface Design Guidelines

Focus

The interface should help user stay focused and give the relevant options and data first

Fitt's Law

Law—"The time to acquire a target is a function of the distance to the target and the size of the target ."

Latency Reduction

Interface must do multitasking rather than kept user waiting for an operation to complete
e.g., display a progress bar to show the progress of the task like downloading or Show and animated visuals to keep user focused.

Revised Interface Design Guidelines for a Web App

Learnability

- Minimize learning time by creating simple & intuitive designs

Maintain work product integrity

- An interface should autosaved the user data
- e.g., filling out long forms and losing data in case of an error in an input field. This leads to frustration. So the data should be saved for all the correct fields.

Readability

- All information presented should be readable by all age group users

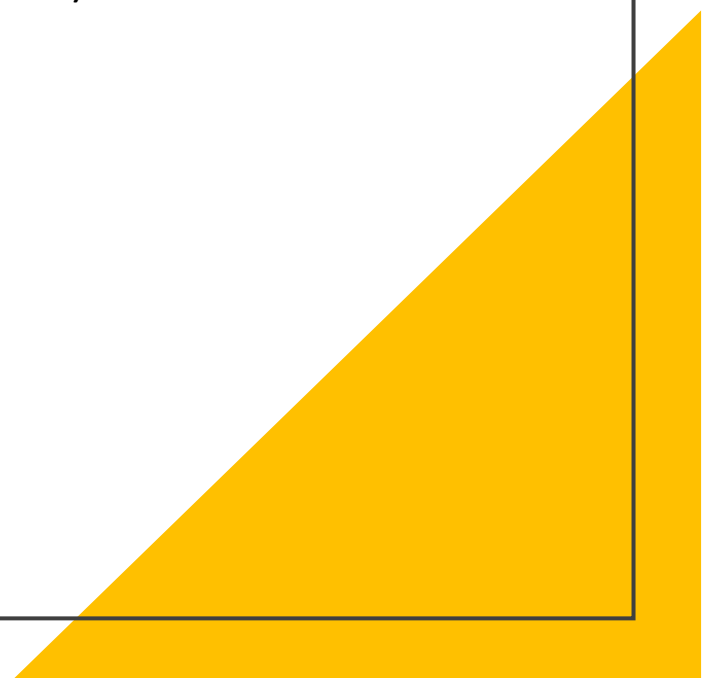
Track state

- Cookies can be used to track user activities so user may logout any time & can continue from the same state after logging in again

Visible navigation

- Obvious navigation bars mentioned on every interface of the application.

Interface Design Workflow:

- Derive and refine information from requirements
 - Develop rough layout sketch
 - Map user objective to actions (like Home, Contact, booking, facilities, etc....)
 - Create storyboard screen images for every interface
 - Refine layouts by improving aesthetics
 - Develop an activity diagram to show the design flow
 - Develop state diagrams to show changes in state after transitions
 - Describe interface layout for every state
 - Refine & review the model.
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Interface Design Evolution

